

Wobbly Liquids (and more) [draft 1]

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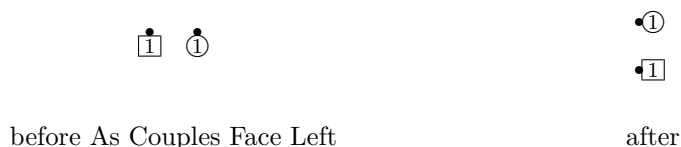
1 Introduction

Here are some concepts in modern western square dancing.

- **As Couples:** do the call with each dancer replaced with two adjacent dancers, who turn as one unit.
- **Couples Twosome:** do the call with each dancer replaced with two adjacent dancers, who turn individually.
- **Stable:** do the call without turning.

The first two concepts seem unrelated to the last one. But the relationship between As Couples [call] and Couples Twosome [call] actually has a similarity to the relationship between [call] and Stable [call].

When dancing a call As Couples, consider what happens when a virtual dancer¹ turns 90° to the left. The two real dancers both individually turn 90° to the left, but they also change positions on the floor: they *revolve* 90° around their midpoint.



Now consider Couples Twosome instead. When a virtual dancer turns 90° to the left, the two real dancers still both individually turn 90° to the left. The only difference is that they do *not* change positions on the floor; in other words, they revolve 0° around their midpoint.



How does this relate to Stable? Suppose we defined the pointless concept “Normally,” which has no effect (to do a Normally Scoot Back, simply do a Scoot Back). Then, to dance a call Normally, when a virtual dancer turns 90° to the left, the real dancer turns 90° to the left. But to dance a call Stable, when a virtual dancer turns 90° to the left, the real dancer does *not* turn; in other words, they turn 0° to the left.

¹To clarify some terminology, suppose we are considering the call As Couples Scoot Back, which takes 8 dancers. I will call Scoot Back, the “virtual call,” and there are 4 “virtual dancers” (each corresponding to two “real dancers” in the “real call.”)

I will use the same terminology for all concepts – to dance a Stable Scoot Back, the real dancers follow the same path on the floor as the virtual dancers, but the virtual dancers turn and the real dancers do not.

We can summarize this parallelism in a table.

	[virtual 90° rotation] maps to [real...	
	...90° ...	...0° ...
...rotation]	(Normally)	Stable
...revolution]	As Couples	Couples Twosome

This is surprising! Here’s a different way of thinking about it: every Solid concept (like As Couples or Tandem) contains a “revolution component,” where the two dancers are moving on the floor relative to each other even if the base call only turns in place. Twosome basically makes this revolution component “Stable.”

2 Two new concepts

That table is cool, but there’s no reason to stop there. What’s so special about 90° and 0°?

	[virtual 90° rotation] maps to [real...		
	...90° ...	...0° ...	...−90° ...
...rotation]	(Normally)	Stable	???
...revolution]	As Couples	Couples Twosome	???

2.1 Wobbly

The first cell I’ll fill in is the top right one. What exactly does this combination mean?

Unfolding the layers of indirection, it says “virtual 90° rotation maps to real −90° rotation.” In simpler words, you **reverse the direction of rotation**: if the base call turns left 90°, then you turn right 90°.

Wobbly [call]: Walk the same path on the floor that you would for [call]; but whenever you would turn left, turn right instead, and vice versa.

For example, Reverse Explode could be defined as Wobbly Explode.

2.2 Liquid

What about the bottom right cell? This one says “virtual 90° rotation maps to real −90° revolution.”



before Couples are Liquid Face Left



after

This movement is a bit harder to explain than either As Couples or Couples Twosome. You could technically write down a definition, which I have done below, but really I think you should try to develop an intuitive sense for what the motion is like.

[group] are Liquid, [call]: Treat each [group] as a single dancer in [call]; but whenever you turn a given direction, revolve each dancer in the [group] around its center point in the *opposite* direction by an equal amount.

For example, from lines facing out, Explode is the same as Couples are Liquid Face In.

We can slot these into the table:

	[virtual 90° rotation] maps to [real...		
	...90° ...	...0° ...	...-90° ...
...rotation]	(Normally)	Stable	Wobbly
...revolution]	Solid	Twosome	Liquid

I have amended the bottom row slightly, so as to not mention Couples specifically – the generalized form of As Couples and Tandem is Solid (e.g. “the Boxes are Solid”). The table explains why I chose these names for the two concepts; Wobbly is sort of the “opposite” of Stable, and Liquid is sort of the “opposite” of Solid.²

3 Infinitely many new concepts

That table is cool, but there’s no reason to stop there. What’s so special about 90° and 0° and –90°?

Okay, I can’t write down a table with infinitely many columns. Instead, I will define two parametrized concepts.

[N]-way Stable, [call]: Walk the same path on the floor that you would for [call]; but whenever you would turn by some amount, turn by that amount multiplied by [N] instead.

[group] are [N]-way Twosome, [call]: Treat each [group] as a single dancer in [call]; but whenever you would turn by some amount, revolve each dancer in the [group] around its center point by that amount multiplied by [N].

Every concept mentioned so far can be defined in terms of these concepts.

Normally	Stable	Wobbly
=	=	=
1-way Stable	0-way Stable	-1-way Stable
Solid	Twosome	Liquid
=	=	=
1-way Twosome	0-way Twosome	-1-way Twosome

Here are a few examples.



before 2-way Stable, Pass the Ocean

after (“Pass the Ocean, Roll Like You Could”)



before 1/3-way Couples Twosome, Single Shakedown

after (seems familiar?)

²Don’t think about this sentence too hard.

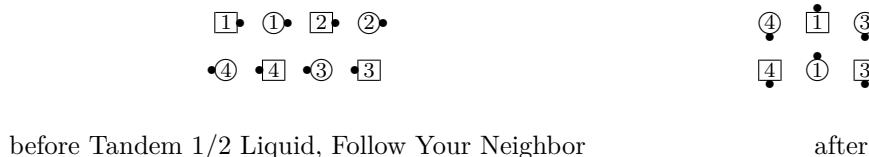
4 Fractional variants

Why did I add the suffix “-way” to these new concepts? They would otherwise collide with existing concept names – “1/2 Stable” and “1/4 Twosome” already mean something!

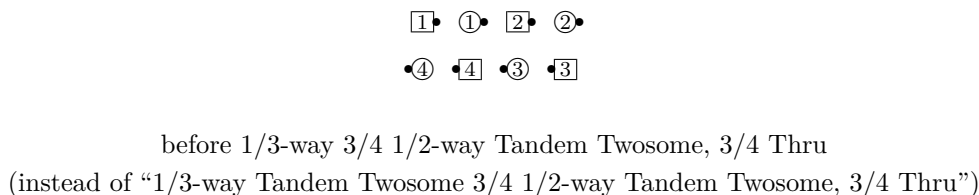
On that note, it’s not a coincidence that both of these names are already used. There is an exact parallelism in the existing concepts:

- “Solid 1/4 Twosome” means “do the call Solid until turning 1/4, then Twosome.”
 - “1/4 Stable” means “do the call Normally until turning 1/4, then Stable.”
- (The “Normally” is simply not pronounced.)

Of course, we can also use fractions with the new concepts!



For the generalized versions, I think the following abbreviated phrasing should be used to avoid excessive repetition (and hopefully sound slightly less confusing).



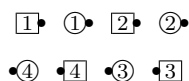
Note that you should be *extremely* careful with fractional applications of generalized Stable. If the initial parameter is not 1 or -1 , dancers will be forced to keep track of a virtually-turned amount which is different from the amount they have actually turned – all while keeping track of a virtual facing direction which is different from their actual facing direction.³ I recommend simply not calling it.

³Also, if the final parameter is 1, the phrasing can get very confusing (“Wobbly 1/4 Peel to a Diamond”???). I recommend explicitly saying “Wobbly 1/4 Normally, Peel to a Diamond,” or even “ -1 -way 1/4 1-way Stable, Peel to a Diamond.”

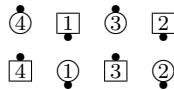
5 “Dynamic” variants

There is an interesting concept Dynamic,⁴ not currently in use, due to Leon Eskenazi:

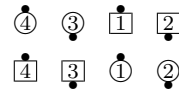
[group] are Dynamic, [call]: Treat each [group] as a single dancer in [call]; but work Solid when turning right, and Twosome when turning left.



before Tandem Dynamic, 1/4 Thru



halfway (working Solid)



after (working Twosome)

Using the terminology from this paper, we could say that Dynamic is “(1 when turning right, 0 when turning left)-way Twosome.” But again, what’s so special about these numbers?

[group] are [N]-and-[M]-way Twosome, [call]: Treat each [group] as a single dancer in [call]; but whenever you would turn by some amount, revolve each dancer in the [group] around its center point by that amount multiplied by [N] if turning right, and [M] if turning left.

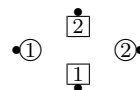
Then Dynamic is 1-and-0-way Twosome. Of course, the same thing works for generalized Stable, although tracking a virtual facing direction while turning can get confusing.



before 0-and-(-1)-way Stable,
Beep Beep



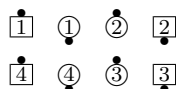
halfway



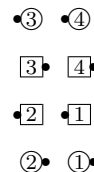
after

6 Melded variants

The Melded Solid/Twosome concept generalizes in an extremely natural way to N-way Twosome, almost so much so that I don’t feel the need to explicitly mention it. But I figured I should at least provide an example.



before Melded Couples are Liquid, Single Checkmate



after

Similarly, one could call Phantom Tandems are Liquid, Gruesome Couples are Liquid, and so on. The 3x1 family of concepts, which is frequently combined with Solid and Twosome, also works particularly nicely.

⁴Full paper: <https://challengedance.org/dynamic.pdf>

7 Examples

The flexibility of these concepts allow for some fun combinations. In the following table, the base call is the same for all cells. (The concept on top comes first, then the concept on the left.)

	“Normally”	Stable	Wobbly
Couples are Solid	Lead Left	Belles Cross	(Mimic Ends Tunnel In)
Couples Twosome	(1/4 Stable Turn to a Line)	Veer Left	(—) ⁵
Couples are Liquid	Reverse Split Swap	Beau Walk Belle Dodge	Circle Left 1/4

These concepts can also be used to describe some “cheats” that I enjoy. For example, the Leads in a Follow Your Neighbor or the Belles in a Mini-Chase can cheat by working $-1/3$ -way Stable. Other calls can be danced Wobbly just for fun, like Column Circulate as #1.⁶ On Square Thru, dancing -3 -way Stable is particularly fun.

Here are a few miscellaneous equivalences not yet mentioned.

Cross Lockit	=	Couples are Liquid, Hinge (from 2fl)
Scout Apart	=	$-1/2$ -way Tandem Twosome, Trade
Travel Thru	=	Belles are Wobbly, Pass the Ocean
Chase the Tag [N/4]	=	Belles are Wobbly, Vertical Tag [N/4]
Swap Around	=	Wobbly, Mimic Centers of Columns, In Style

8 A cool mathematical theorem

Let’s go back to the table at the top of section 7. I mentioned the intended concept order. Does the order matter?

Yes – **Wobbly, Couples are Liquid**, Single Lead Left is a Circle Left 1/4; but **Couples are Liquid, Wobbly**, Single Lead Left is something like a Mimic Ends Tunnel In.

Wait. That’s the same as the cell for **Wobbly, Couples are Solid** above. Is that a coincidence? If you try a bunch more examples, you might realize that you find many more cases where **Liquid, Wobbly** is the same as **Wobbly, Solid**.

Is this always true? As it turns out, yes; in fact, there is a more general theorem about the relationship between these two concepts.

Theorem 1. *For any n , m , and $[call]$, the following are equivalent:*

- n -way Twosome m -way Stable $[call]$
- m -way Stable nm -way Twosome $[call]$

Proof. Compute the first formulation. Consider m -way Stable $[call]$ – this has individual rotation scaled by m . Now apply n -way Twosome to the result. This still has individual rotation scaled by m , but revolution scaled by $m \cdot n = nm$ with respect to the original call (since the rotation of the call it sees is already scaled by m with respect to the original call).

Now compute the second formulation. Consider nm -way Twosome $[call]$ – this has individual rotation scaled by 1 and revolution scaled by nm . Now apply m -way Stable to the result. This has individual rotation scaled by $1 \cdot m = m$, and still has revolution scaled by nm . $\square$

⁵There isn’t really a good name for this, but I guess it’s “Finally Emulate Veer & Turn $1/4 \times 1/4$ ” (with appropriate flow), or “Finally Emulate $2/3$ Finish Clean Sweep $1/4$.” Let me know if you come up with something better...

⁶In some sense, generalized Stable is the “wrong” concept, and the relevant one is “turn some amount in the wrong direction, such that the facing direction you end up with is the same as it would have been if you did the call normally.” This is always generalized Stable with some parameter, but the parameter depends on the call.

Corollary 1. *For any n , m , and $[call]$, the following are equivalent:*

- *n -way Stable m -way Twosome $[call]$*
- *m/n -way Twosome n -way Stable $[call]$*

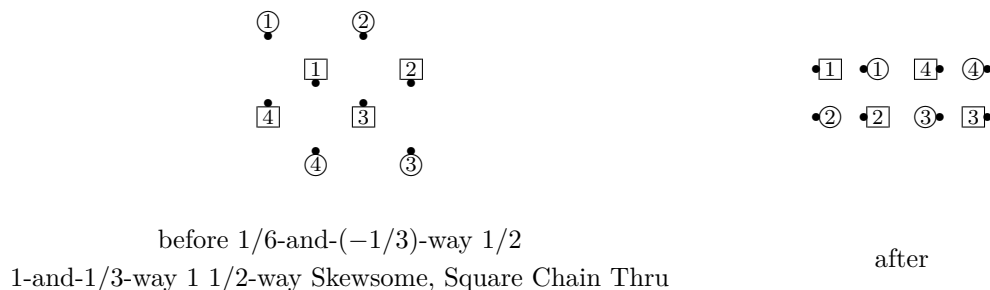
In other words, when swapping the order of a generalized Stable and a generalized Twosome, you must adjust the coefficient on the Twosome multiplicatively by the coefficient on the Stable. When moving the Stable closer to the base call, you divide by it; and when moving the Stable further from the base call, you multiply by it.

One way to see this is to observe that you can never move a 0-way Stable past a (non-0-way) Twosome, since the Stable will then delete the turning information from the original call, making it impossible for the Twosome to determine how much to revolve by. This corresponds to the division by zero that would occur from attempting to apply Corollary 1.⁷

9 Conclusion

The multi-way Stable and Twosome concepts are extremely general and high-powered. But as we all know, with great power comes great responsibility.

It is quite easy to construct wildly incomprehensible calls out of even a single concept application. (Please never call the below example.)



If I keep describing things and telling you not to call them, why am I even writing this paper? Well, the honest answer is that basically none of the motivation came from wanting the concept to be called. Indeed, the main reason I wrote this was to have something to send people when referencing the idea,⁸ instead of having to re-explaining the same thing over and over.

But in some sense, I was “forced” to define bad concepts, because it just wouldn’t make any sense to leave them out. Fractional Liquid is a fine application, so I wanted to define it. But then fractional Wobbly practically defines itself, and leaving an inexplicable hole in the otherwise nicely structured array of concepts would be completely illogical. It is simply an unlucky coincidence that fractional Wobbly, to use the technical term, sucks.

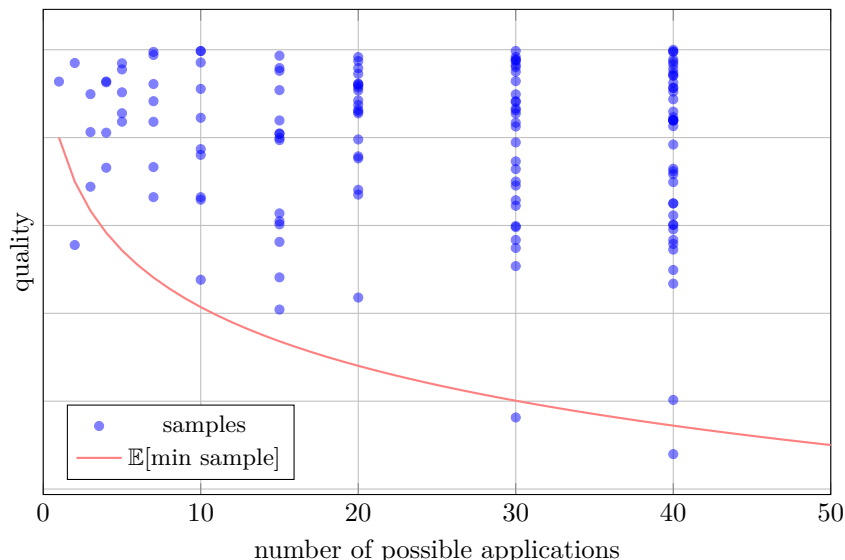
I decided to go off on a billion tangents about this notion of how general a concept is, in relation to the floor of its quality, in the “Excessively verbose conclusion” below, in case reading that is your idea of fun. In any case, I hope this paper has served its primary purpose of both being entertaining to read, and inducing the mind-bending puzzling over abstractions so characteristic of this absurd activity.

⁷As a point of amusement, from this corollary we can derive the identity $0/0 = 0$.

⁸Which has come up organically many times, believe it or not.

10 Excessively verbose conclusion

As just acknowledged, this paper has a concerning density of statements of the form “please don’t call this.” Is this a flaw with the new concepts? Personally, I don’t think so. My opinion is that this is just a direct consequence of the generality of the concepts. More specifically, there is a rough inverse relationship between the level of expressiveness of a tool in square dancing and the worst thing you can do with it, or in other words the level of care a caller must take when using it.



The model depicted here is a huge approximation, but I think it contains some kernel of truth. The x-axis is the “number” of things you can do with the concept, by some metric – as you move further to the right, there are more points on the graph per x-coordinate. The y-axis represents the “quality” of what happens if you try each thing.

Of course, some things will work better than others, and we can imagine the quality of each thing is drawn randomly from the same distribution. Hopefully, this distribution is heavily left-skewed, since the concepts we choose are the ones that usually result in good calls. But as you take more samples from the distribution, the expected minimum (i.e. the worst call you get) decreases.

As a more concrete example, consider where the following two “tools” would land on this graph:

1. you’re only allowed to call Swing Thru
2. you’re allowed to call Swing Thru, Swing, Slip, and Acey Deucey

The only thing you can do with (1) is call Swing Thru repeatedly, which is not very interesting but a fine thing to do. So you can imagine it corresponds to one medium-height point on the graph.

You get more expressive power with (2), which results in some higher points corresponding to more interesting things you can do (say, Acey Deucey followed by Swing Thru). But this expressivity also allows you to do some much worse things, such as “Swing Thru, Acey Deucey, Slip” (which sucks for the new centers). So the lowest point on the graph is much lower.

Ok, that was some fun rambling about generalities, but all that being said... here is my personal summary of what I think the value of this particular writeup is.

- As far as actual dancing goes, simple applications of these concepts are probably fine, with some workshopping.

- The introduction of the language of these concepts is very helpful for talking about square dancing, as in section 7; e.g. to explain the relationship between Lead Right and Split Swap, or to describe a dance action like the Follow Your Neighbor cheat.
- Writing the concepts down lets us analyze them mathematically, and prove cool facts like Theorem 1.
- At some level, there is an intrinsic value in generalization. It reveals relationships between concepts, instead of forcing us to think separately about the Stable, Solid, Twosome, Fractional Twosome, Dynamic, etc. concepts; and having to reinvent different combinations as we think of new ones like Liquid.
- In my own opinion, the “un-danceable” applications are very fun to think about, and I would personally have fun if they were called (but again, do not).